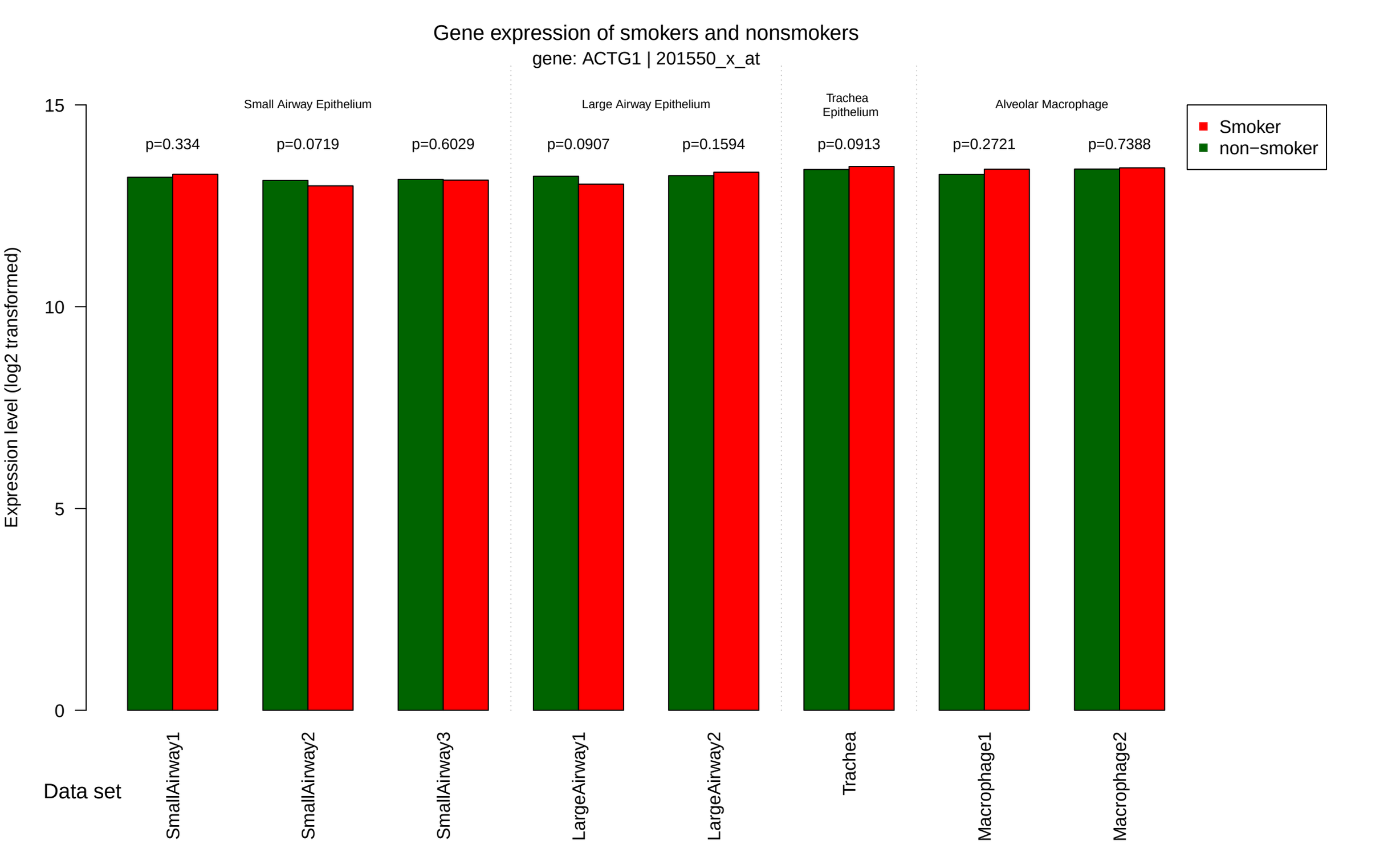
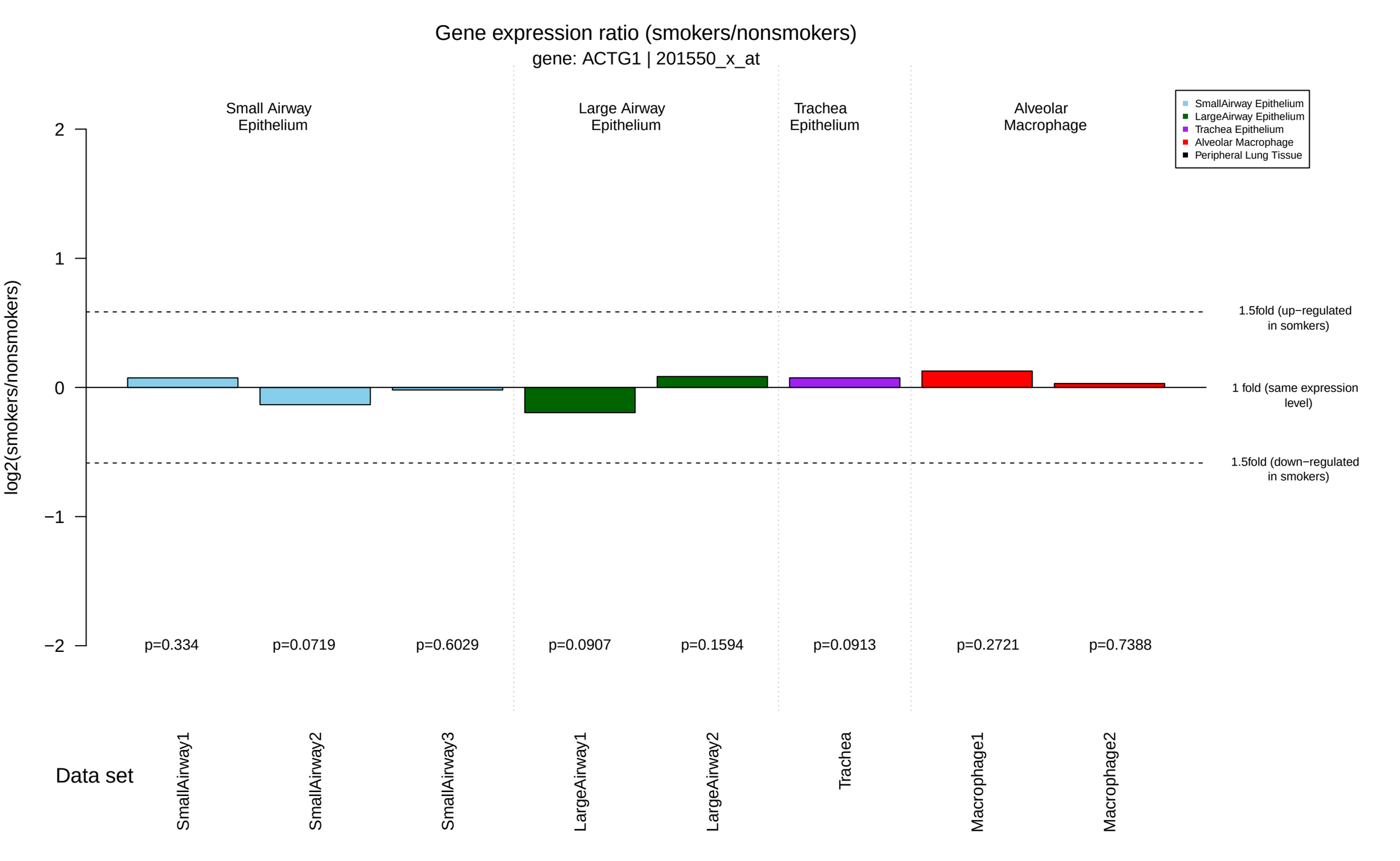
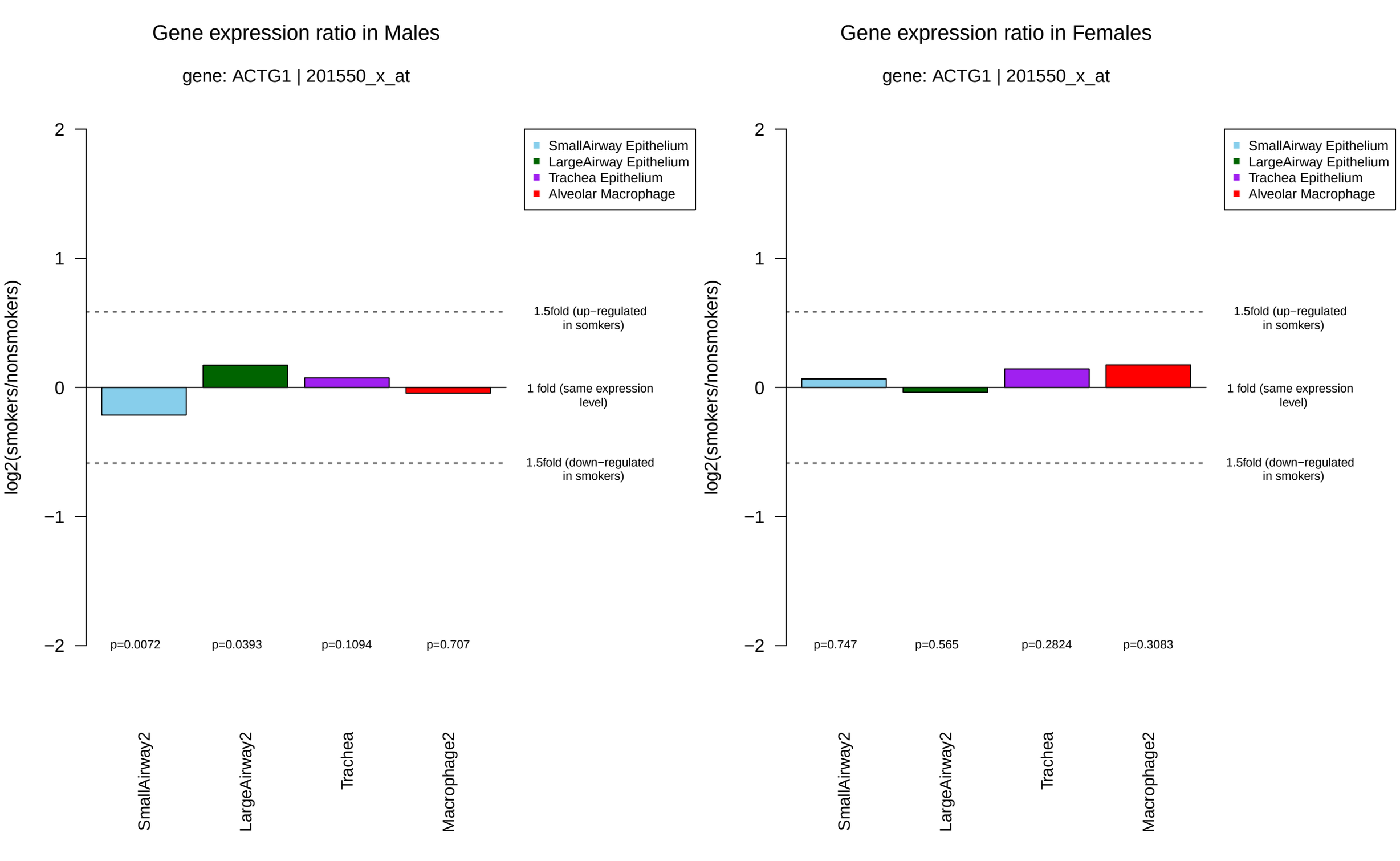
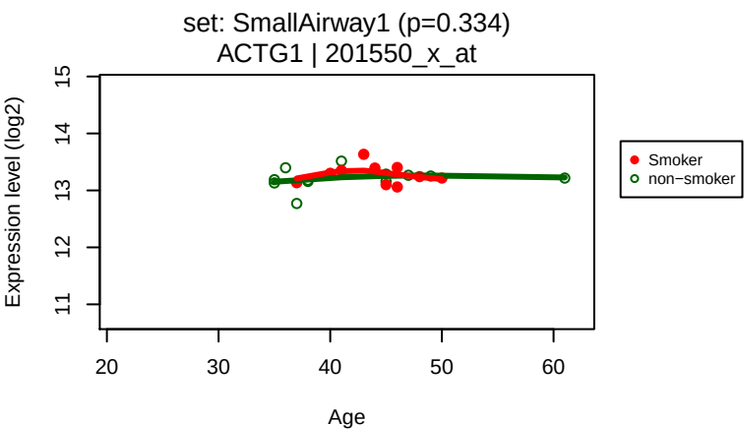
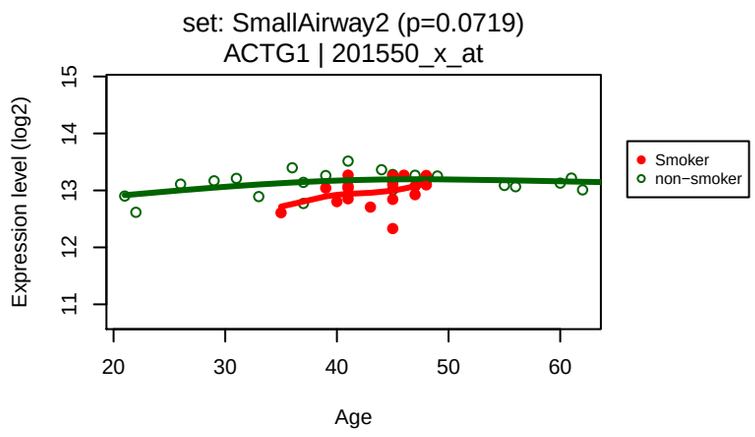
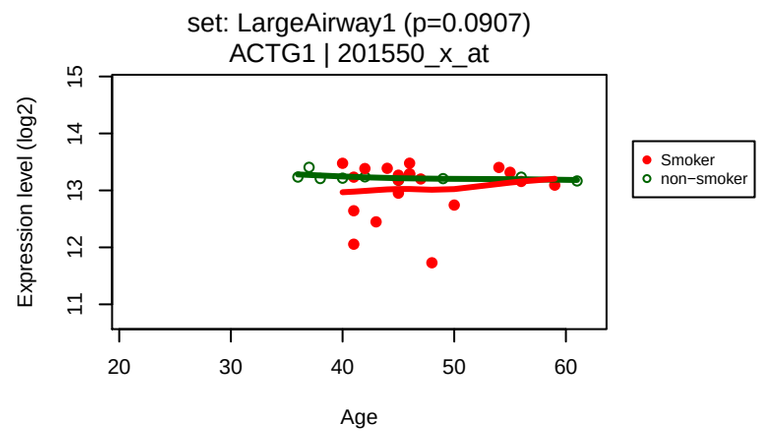

Gene expression of smokers and nonsmokers
gene: ACTG1 | 201550_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.334
p=0.0719
p=0.6029
p=0.0907
p=0.1594
p=0.0913
p=0.2721
p=0.7388
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

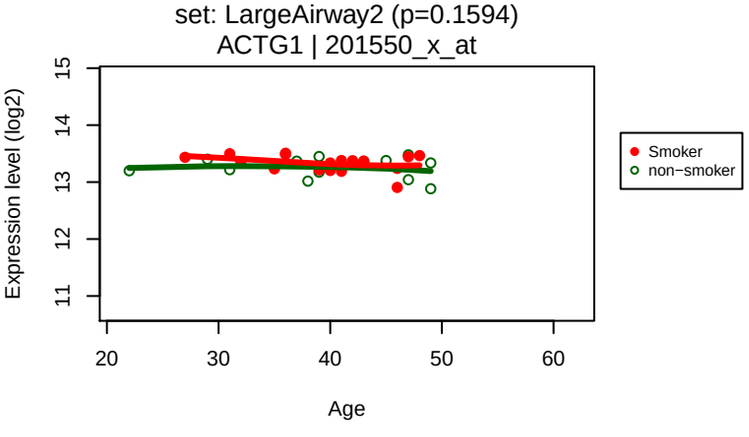

Gene expression ratio (smokers/nonsmokers)
gene: ACTG1 | 201550_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.334
p=0.0719
p=0.6029
p=0.0907
p=0.1594
p=0.0913
p=0.2721
p=0.7388
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

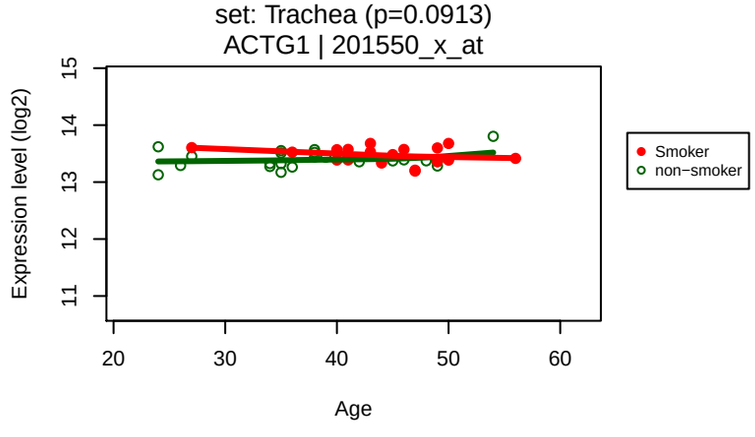

Gene expression ratio in Males
gene: ACTG1 | 201550_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0072
p=0.0393
p=0.1094
p=0.707
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACTG1 | 201550_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.747
p=0.565
p=0.2824
p=0.3083
SmallAirway2
LargeAirway2
Trachea
Macrophage2

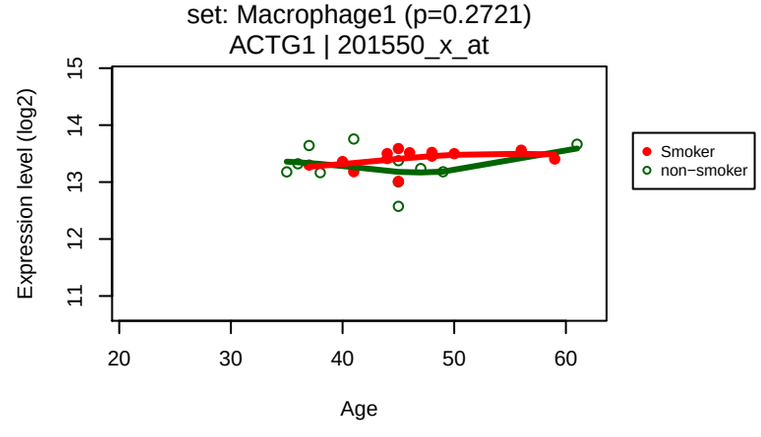

set: SmallAirway1 (p=0.334)
ACTG1 | 201550_x_at
Expression level (log2)
Age
Smoker
non−smoker

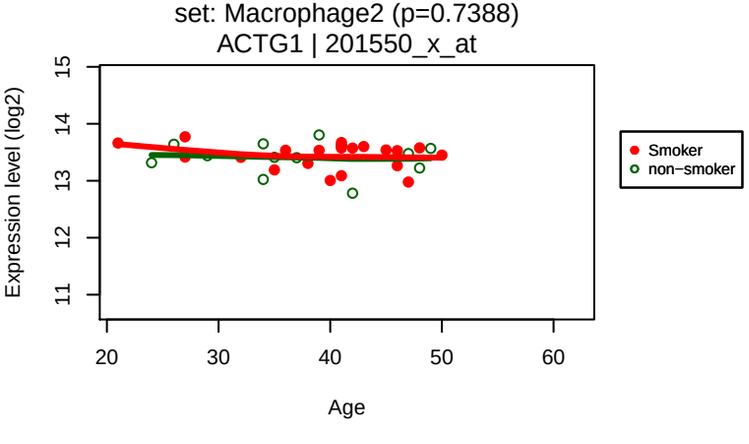

set: SmallAirway2 (p=0.0719)
ACTG1 | 201550_x_at
Expression level (log2)
Age
Smoker
non−smoker

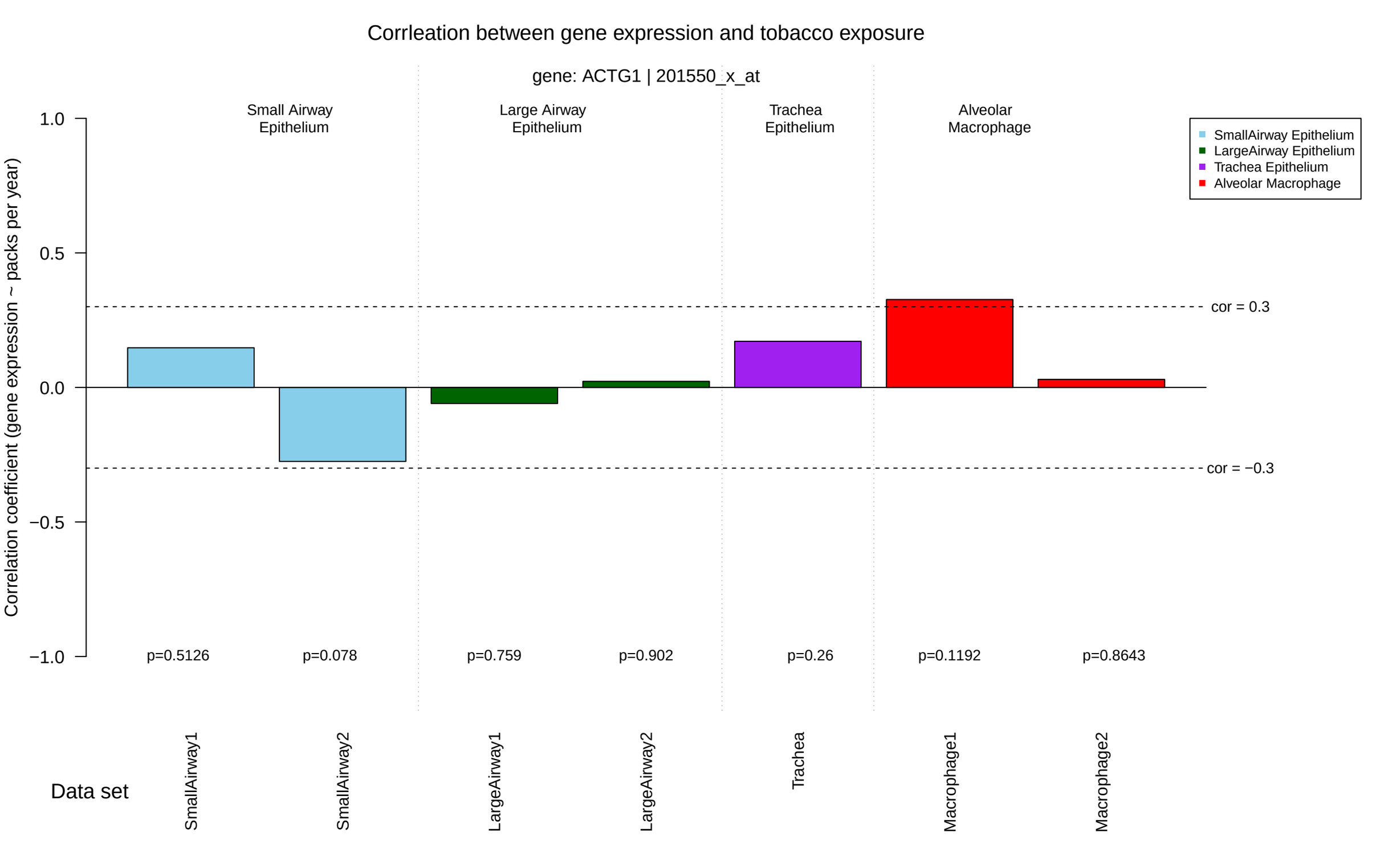

set: LargeAirway1 (p=0.0907)
ACTG1 | 201550_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1594)
ACTG1 | 201550_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0913)
ACTG1 | 201550_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2721)
ACTG1 | 201550_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7388)
ACTG1 | 201550_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACTG1 | 201550_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5126
p=0.078
p=0.759
p=0.902
p=0.26
p=0.1192
p=0.8643
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2